

JIAPENG ZHANG 张嘉鹏

✉ jiapengzhang@mails.tsinghua.edu.cn · 🌐 zjp-shadow.github.io · 🔗 zjp-shadow · 📄 Google Scholar

🎓 EDUCATION

Tsinghua University, Beijing, China

Sep. 2024 – Present

Ph.D. Student in Department of Computer Science and Technology

- **GPA:** 3.95 **Rank:** 27/158
- **Advisor:** Prof. Shi-Min Hu (Institute of Human-Computer Interaction and Media Integration)
- **Research Focus:** 3D Generation, Animation, Graphics Simulation & Embodied AI, and AI Infra.

Tsinghua University, Beijing, China

Sep. 2020 – Jul. 2024

B.S. in Information and Computing Science, Zhili College

- **Rank:** 21/30

📖 SELECTED PUBLICATIONS

• SkinTokens: A Learned Compact Representation for Unified Autoregressive Rigging

Jiapeng Zhang, Chengfeng Pu, Menghao Guo, Yan-Pei Cao, Shi-Min Hu

ACM Transactions on Graphics (TOG), 2026, *Accepted with Minor Revisions*

(zjp-shadow.github.io/works/SkinTokens/)

Introduced *SkinTokens*, a learned discrete representation for skinning weights. By framing skinning as a sequence prediction task, the unified *TokenRig* framework jointly models skeletons and skin deformations, achieving substantial improvements in rigorous out-of-distribution geometry evaluations.

• One Model to Rig Them All: Diverse Skeleton Rigging with UniRig

Jiapeng Zhang, Chengfeng Pu, Menghao Guo, Yan-Pei Cao, Shi-Min Hu

ACM Transactions on Graphics (TOG) (SIGGRAPH 2025) (zjp-shadow.github.io/works/UniRig/)

Proposed UniRig, a unified framework for automated skeletal rigging leveraging large autoregressive models and a novel *Skeleton Tree Tokenization* method. Trained on *Rig-XL* (a new large-scale dataset), it efficiently generates topologically valid structures, far outperforming existing state-of-the-art tools across diverse generic models.

• CharacterGen: Efficient 3D Character Generation from Single Images with Multi-View Pose Canonicalization

Haoyang Peng, Jiapeng Zhang, Menghao Guo, Yan-Pei Cao, Shi-Min Hu

ACM Transactions on Graphics (TOG) (SIGGRAPH 2024), 43(4): 1–13, 2024 (charactergen.github.io/)

Presented a highly efficient framework for lifting single images into pose-unified 3D character meshes. It utilizes an image-conditioned multi-view diffusion model for pose calibration and a transformer-based sparse-view network, producing geometrically and texturally robust outputs ready for animation.

• Transferring the Intelligence of VLMs to Robotic Control

Menghao Guo, Zhehan Mo, Jiajun Wang, Yi Zhang, Kejin Wang, Yixuan Deng, Jiapeng Zhang, Yongming Rao, Shi-Min Hu

arXiv preprint arXiv:2609.22966, 2026 (robodawn.top/)

Introduced *RoboDawn*, a human-intuitive interface that lets an agentic VLM control robots in a closed loop through discrete translation, rotation and gripper commands, grounded by a few in-context demonstrations. Without task-specific robot training it achieves state-of-the-art success rates on RoboTwin 2.0 C2R and RoboDojo, and transfers to a real Franka arm.

🔗 OPEN SOURCE PROJECTS

UniRig & SkinTokens (github.com/VAST-AI-Research/UniRig, SkinTokens)

2200+ Stars

Project Maintainer & Core Contributor

- Maintain the official open-source implementations of UniRig (SIGGRAPH 2025, 1.8k+ stars) and SkinTokens (TOG 2026, 450+ stars), distributing unified pre-trained skeleton and skinning models that rig multi-genre meshes, from functional humans to fictional inorganic artifacts.
- Released the curated *Rig-XL* dataset and the unified *TokenRig* pipeline that predicts skeletons and skinning weights in a single autoregressive model, enriching the broader 3D automation community.

CharacterGen (github.com/zjp-shadow/CharacterGen)

830+ Stars

Project Maintainer & Core Contributor

- Designed and actively maintained the rendering and generation inference tools driving single-image 3D lifting, resolving pose ambiguity with canonical “A-pose” generation.
- Open-sourced multi-view diffusion synthesis and detailed texture back-projection, widely adopted by 3D anime avatar creators.

Tsinghua Jittor Framework (github.com/Jittor/Jittor, JittorInfer)

3200+ Stars

Core Member & Head of JittorInfer Framework

- Engineered **JittorInfer**, a high-performance C++ LLM inference framework adapted to domestic AI hardware such as Huawei Ascend processors.
- Spearheaded system-level computing abstractions, including CPU multi-stage operator fusion and end-to-end resource execution scheduling, which brought large throughput improvements when serving frontier models like the **DeepSeek** lineup compared to Ascend-vLLM.

 HONORS AND AWARDS

<i>Rising Star Pioneer Award, Huawei Kunpeng & Ascend Developer Conference (KADC)</i>	2026
<i>Comprehensive Scholarship for Ph.D. Students, Tsinghua University</i>	2025
<i>Silver Medal, National Olympiad in Informatics (NOI)</i>	2019
<i>Gold Medal, China Team Selection for Informatics (CTS)</i>	2019

 INTERNSHIP EXPERIENCE

Tencent Hunyuan

2026

Research Intern, Video Generation (Tsinghua Summer Social Practice)

- Studied how the metrics of video VAEs relate to the generation performance of flow-based video models.

Tencent

2024

Research Intern

- Conducted research on 3D multi-modal understanding.

Huawei

2023

Software Engineering Intern

- Developed and implemented system-level optimizations for sparse matrix multiplication targeting the SME (Scalable Matrix Extension) instruction set on Kunpeng chips.